

RESEARCH

Open Access



Lightweight CNN-based edge computing with hybrid chaotic encryption for secure and efficient image transmission in IoT networks

Shahnwaz Afzal¹, Mohammad Ubaidullah Bokhari¹, Mohammad Luqman^{1*} and Basil Hanafi²

*Correspondence:

Mohammad Luqman

mdlqman@myamu.ac.in

¹Department of Computer Science,
Aligarh Muslim University, Aligarh,
U.P 202002, India

²School of Computer Science and
Engineering, Galgotias University,
Greater Noida, India

Abstract

While IoT Devices are used in Governmental organizations to send data to Cloud services, there are several Issues with conventional communication methods used to connect devices.1) The secure transmission of confidential Images is not guaranteed.2) Limitations of Memory, Computation, and Bandwidth are present on IoT devices. This Paper presents the development of a framework for the secure transmission of confidential images in real-time in Indian Government IoT Environments using a CNN-assisted Selective Encryption Framework. An Edge-based CNN, which consumes minimal processing power to classify images into either Sensitive or Non-sensitive, is employed to perform automatic classification of digital images. Once classified by the CNN, the Sensitive images will be transmitted using a Chaotic Hybrid encryption scheme in conjunction with the SHA-512 algorithm, thus ensuring Confidentiality and Integrity. Conversely, non-sensitive images will use no Encryption, allowing for a reduction in both processing overhead and service Latency. As a result, the experimental results show that the classification results achieved high accuracy (Training: 98.6%, Validation: 97.2%, Testing: 96.8%) and that the Encryption Algorithms employed in this framework demonstrated state-of-the-art cryptographic security through the analysis of Ciphertext Entropy (7.9972), Correlation (Minimal) and NPCR/UACI values demonstrating high resistance to statistical and differential attacks. Through comparisons made against existing encryption schemes, the Chaotic Hybrid Encryption approach used in this Paper demonstrated an approximate 6.23 times faster throughput than the existing schemes, and a 6.22 times reduction in overall computational Latency, thus presenting new ways of transmitting secure messages through Government IoT Communication in real-time.

Keywords Deep learning, Logistic map, Henon map, IoT, Edge computing, Histogram analysis



1 Introduction

Actuators and sensors enable physical devices compatible with the Internet of Things (IoT) to interact and communicate data. This integrated ecosystem encompasses healthcare, smart homes, industrial automation, and government services, facilitating real-time monitoring, automation, and decision-making guidance [1, 2]. Government infrastructure increasingly employs IoT devices to collect and transmit substantial quantities of image-based data, including financial credentials, identification proofs, PAN cards, and Aadhaar cards, which contain sensitive personal information. The surge in image data poses considerable challenges for scalable, secure, and efficient processing. Government offices have traditionally relied on manual data processing; however, the Internet of Things has facilitated rapid digitalization and service automation. This highlights the importance of assurances regarding data integrity and confidentiality. Nevertheless, contemporary IoT systems possess inherent limitations. The resource constraints, including limited computational capacity, memory, and energy, of most IoT devices render robust encryption approaches inapplicable to all data universally. Furthermore, the perpetual encryption of all data, irrespective of its sensitivity, incurs additional expenses and diminishes system performance. We have examined lightweight cryptographic algorithms such as AES [3], Grain-128 [4], Grain-80 [5], Lizard [6], Rabbit [7], Trivium [8], and Salsa20 [9], notwithstanding the absence of dynamic alterations in data sensitivity or processing constraints at the edge. However, traditional algorithms were not designed for image encryption, so an inquiry was conducted into many supplementary methodologies specifically designed for image security. Their significant processing complexity makes them unsuitable for real-time image applications. DNA-based image encryption algorithms offer dual-layer security; nonetheless, they necessitate considerable computational power. Watermarking and cellular automaton-based lightweight encryption models exist; however, they are either non-scalable or produce image distortion. Some research has proposed different medical picture encryption solutions [10]. present a chaos-based encryption technique employing diffusion from a logistic map. Although this method improved randomization, its deterministic nature left it vulnerable to known plaintext attacks, limiting its practical security. Examining DNA cryptography for medical imaging security [11], achieved high-security levels, but at the cost of significant computational overhead, rendering real-time applications impractical [11]. Arnold's Cat Map was utilized to obfuscate medical images before encryption, significantly reducing the association between neighboring pixels. An urgent requirement exists for a lightweight, secure, and reversible encryption technique that ensures the robust security of sensitive images without depleting device resources or affecting performance.

Encrypting all types of images with a single algorithm imposes a significant computational burden on edge devices, resulting in inefficiencies in processing time, energy consumption, and overall system responsiveness. In the context of the proposed model, our research effectively addresses this challenge by introducing a novel architecture that integrates IoT, edge computing, and deep learning to provide sensitivity-aware image encryption, thereby overcoming these limitations. This research is motivated by the increasing adoption of IoT in governmental services, where the secure handling of sensitive images is essential. Implementing content-aware, real-time security solutions on low-powered IoT devices necessitates minimal latency and robust throughput.

In contrast to other studies, our focus is primarily on ensuring the confidentiality and integrity of data in transit, rather than addressing authentication, authorization, or non-repudiation.

In the proposed methodology, photographs of IoT devices are transmitted securely to the edge layer, where a CNN categorizes each image as either sensitive or non-sensitive. The implementation of a hybrid chaotic encryption model, which integrates logistics and Henon maps, followed by SHA-512 hashing for integrity verification, produces robust pseudorandom sequences suitable for encrypting sensitive photos based on this categorization. Non-sensitive photographs stay unencrypted. Our selective encryption method significantly reduces processing overhead while ensuring security.

The contributions of this paper are as follows:

1. This paper introduces a novel architecture for secure image transmission in government services, integrating chaos-based encryption with edge-based CNN classification, an area inadequately addressed in the literature.
2. The suggested architecture enhances efficiency and minimizes the effort required by IoT devices through real-time edge sensitivity categorization and selective encryption.
3. A lightweight chaotic hybrid model protects sensitive photos; SHA-512 ensures the integrity of received data.
4. Experimental findings indicate a 6.22 times enhancement in throughput and a 6.22 times reduction in encryption time compared to conventional approaches, alongside around 97% classification accuracy, robust entropy, and negligible correlation in encrypted images.

Our study addresses a significant research gap by providing a secure, rapid, and scalable framework specifically designed for government IoT systems, where the secure handling of sensitive images is crucial. The organization of this research is as follows: Sect. 2 presents the related work. Section 3 outlines the preliminaries, while Sect. 4 presents the methodology of the model. Section 5 presents the proposed work, followed by Sect. 6, which discusses the results and conclusions. Sect. 7 discusses the limitation of the study, finally Sect. 8 presents the conclusion and outlines future directions.

2 Related work

Most conventional cryptographic algorithms like AES, DES, and RSA were designed to support encryption/decryption of text and binary files, rather than encryption/decryption of large amounts of high-redundancy images. As a result, they face challenges in efficiently processing and managing the large amounts of pixel correlation, volume of data, and speed at which images must be processed on image-centric. Thus, researchers have turned towards the development of chaos-based, lightweight, and content-adaptive encryption techniques specifically developed to protect images.

2.1 IoT and cloud security challenges

The integration of IoT and cloud services raises numerous questions regarding computational efficiency, data security, and privacy. Due to their intrinsic resource constraints, IoT devices function within computational limitations. Numerous academics have endeavored to address this issue by devising security strategies tailored to the specific requirements of IoT environments. For instance [1], established an ensemble-based

approach for anomaly detection in smart agriculture, whereas [12] introduced an IoT–fog–cloud framework for real-time COVID-19 diagnostics, therefore managing the comprehensive data flow from collection to visualization. In accordance with this [13], created an intelligent IoT gateway utilizing Artificial Neural Networks (ANNs) to identify data flow irregularities, hence improving system security [14]. developed a lightweight classification model for IoT network data using logistic regression and logit boosting, thereby facilitating effective device profiling and security monitoring. These numbers indicate increasing efforts across various levels of security; however, a disregard for selective encryption based on content sensitivity.

2.2 Edge computing for secure image processing

Edge computing facilitates data processing in proximity to its origin, hence reducing latency and bandwidth requirements, two critical elements for real-time IoT devices [15]. introduced a remote patient monitoring system utilizing edge-deployed deep learning models to recognize ECG data based on sensor input [16]. Also had a cloud-specific security level determined by sensitivity according to a data classification system, but it did not support real-time edge deployment. These solutions overlooked CNN-based image classification for selective encryption while showcasing the efficacy of edge computing.

2.3 CNN models in edge

CNNs are increasingly employed in edge devices due to their compatibility with contemporary technology and superior pattern recognition capabilities. A hybrid Adaptive Gated Recurrent Unit–Stacked Convolutional Autoencoder (AGRU–SCAE) model for intrusion detection in smart agriculture, as reported in [17], reveals CNN-like architectures on edge nodes. Furthermore [18], demonstrated Edge Lightweight Spatial–Temporal CNN Pipeline (ELSCP) for anomaly detection in spatial-temporal data streams, hence emphasizing the significance of CNN in IoT applications connected to agriculture. Although their primary emphasis is on anomaly detection rather than image sensitivity classification for encryption selection, these studies demonstrate the feasibility of employing deep models at the edge.

2.4 Chaotic image encryption schemes

Due to their computational simplicity and sensitivity to initial conditions, chaotic systems are frequently employed in photo encryption for lightweight IoT contexts. A hybrid method integrating the Arnold Cat Map and logistic map with AES encryption, enhancing entropy and resistance to differential attacks, is presented in [19]. Similarly [20], introduced a block-wise chaotic encryption technique that iteratively shuffles and applies pseudorandom masking, hence augmenting unpredictability. Opportunities are also evaluated based on DNA. Regarding the encryption of RGB images [21], introduced a hybrid Elliptic Curve Cryptography (ECC)- Deoxyribonucleic Acid (DNA) model, while [22] proposed a dual-permutation scheme suitable for medical imaging [23]. content-aware DNA-based pixel encoding was utilized to resist known-plaintext attacks while [24] proposed a hyperbolic sine function that was devised to enhance robustness against statistical attacks. Hybrid cryptography, as designed given by [25], establishes a dual-layer security architecture with DWT-based watermarking [26]. introduced Intelligent

Encryption Cellular Automaton (IECA), a programmable cellular automaton-based encryption system designed for IoT, whereas [27] proposed an IoT-specific encryption model (IETD) utilizing Tinkerbell and Duffing maps [28]. Chaotic maps were integrated with compressive sensing for effective large-image encryption while [29] proposed an innovative DNA coding and multi-chaotic methods for grayscale image security were presented. While their initial contribution does not pertain to content-aware encryption through edge-classified CNNs, these findings underscore the increasing significance of integrating chaos theory with lightweight encryption [30]. [30] proposed a new multi-image encryption framework that combines a five-dimensional hyperchaotic generator with Arnold's Cat Map and Langton's Ant mechanism to deliver robust security, computational efficiency, and strong resistance to cryptanalytic attacks [31]. Offers a state-of-the-art 3D chaotic-map based encryption with high key space and strong NPCR/UACI metrics which are useful baseline for comparing key-space and security strength [32]. Introduces an encryption scheme with bit-level diffusion and block permutation which shows accepted techniques for strong diffusion & resistance against plaintext attacks [33]. [33] proposed a method through the use of the Logistic Map as a chaotic sequence generator and the use of an enhanced Joseph traversal based on multiple starting points and adaptively sized steps to scramble an image, and a block diffusion either singularly or redundantly rearranges the scrambled image. The enhanced Joseph traversal with the Discrete Fourier Transform can significantly enhance the security of the scrambled image against all known types of cryptanalyses based on standard techniques [34]. Shows application of chaos-based encryption in IoT/medical imaging context, combining hashing and chaotic maps which are relevant for discussion of real-world deployment in IoT.

2.5 Hashing for data integrity

Essentially, secure communication systems rely on data integrity, especially in scenarios where modifying or changing sensitive information could yield adverse consequences [35] and [36]. Guarantees message correctness and non-repudiation by the utilization of Hash-based Message Authentication Code (HMAC) in conjunction with One-Time Password (OTP) methods and encryption. Ideal for edge-layer processing in resource-constrained IoT systems, our method streamlines this by using a SHA-512 hash function solely for data integrity verification, thereby providing robust security without the computational burden of comprehensive authentication systems.

2.6 Limitations and motivation for this study

Numerous unresolved vulnerabilities persist despite significant advancements in IoT security. Regardless of the topic, the majority of previous research either depended only on centralized processing or consistently encrypted all data. Moreover, hybrid encryption systems that combine classical and chaotic ciphers necessitate real-time CNN-based sensitivity assessment to choose the appropriate encryption pathway. Likewise, although SHA-based integrity algorithms are utilized to some extent, they are not designed for implementation in lightweight, edge-deployed systems. Consequently, our research introduces a content-aware, real-time security framework utilizing CNNs at the edge to identify IoT images and employs hybrid chaotic encryption for sensitive images,

thereby providing an innovative, efficient, and secure method specifically designed for government services. SHA-512 pertains to integrity.

3 Preliminaries

This document presents the essential components and theoretical concepts underlying the proposed secure photo classification and encryption technique. These preliminaries facilitate our comprehension of the security, foundation, and framework of our methodology, along with its functionality. We specifically examine CNNs, which serve as the classification engine utilized at the edge; the mathematical formulations and chaotic characteristics of the Logistic and Henon maps employed for key stream generation; the Zigzag scanning technique implemented to enhance pixel diffusion in encrypted images; and the Secure Hash Algorithm 512 (SHA-512), which ensures the integrity of transmitted image data. The proposed approach integrates these components to provide efficient, lightweight, content-aware security suitable for resource-constrained IoT and edge systems.

3.1 Convolutional neural networks (CNN)

CNNs are predominantly intended for image recognition and classification in deep learning models. The suggested approach autonomously identifies incoming images as either sensitive or non-sensitive via an edge layer CNN. The architecture consists of several convolutional layers utilizing ReLU activation functions; batch normalization and max-pooling layers facilitate the extraction of hierarchical information while reducing spatial dimensions. Fully connected dense layers finalize the classification process, while dropout layers mitigate overfitting. Due to its capability for real-time, content-aware decision-making regarding image sensitivity with high precision and computational efficiency, CNN is ideally suited for edge deployment in IoT-driven systems.

3.2 Logistic map

The logistic map, a fundamental yet highly nonlinear mathematical function, is occasionally employed in chaos-based encryption due to its sensitivity to initial conditions and settings. Equation 1 illustrates this point.

$$y_n = \delta y_{n-1} (1 - y_{n-1}) \quad (1)$$

Here, y_{n-1} represents a real number inside the interval $(0, 1)$, while δ denotes a control parameter typically set between 3.57 and 4 to induce chaotic behavior, as shown in Fig. 1. The proposed encryption method generates a pseudorandom byte stream, which is used as a key stream in XOR-based encryption using a logistic map. Its chaotic nature ensures unpredictability and resilience against brute-force attacks, making it suitable for secure photo encryption in lightweight environments.

3.3 Henon map

The Henon map is a two-dimensional discrete-time dynamical system that exhibits chaotic behavior for specific parameter values. It originates from Eqs. 2 and 3.

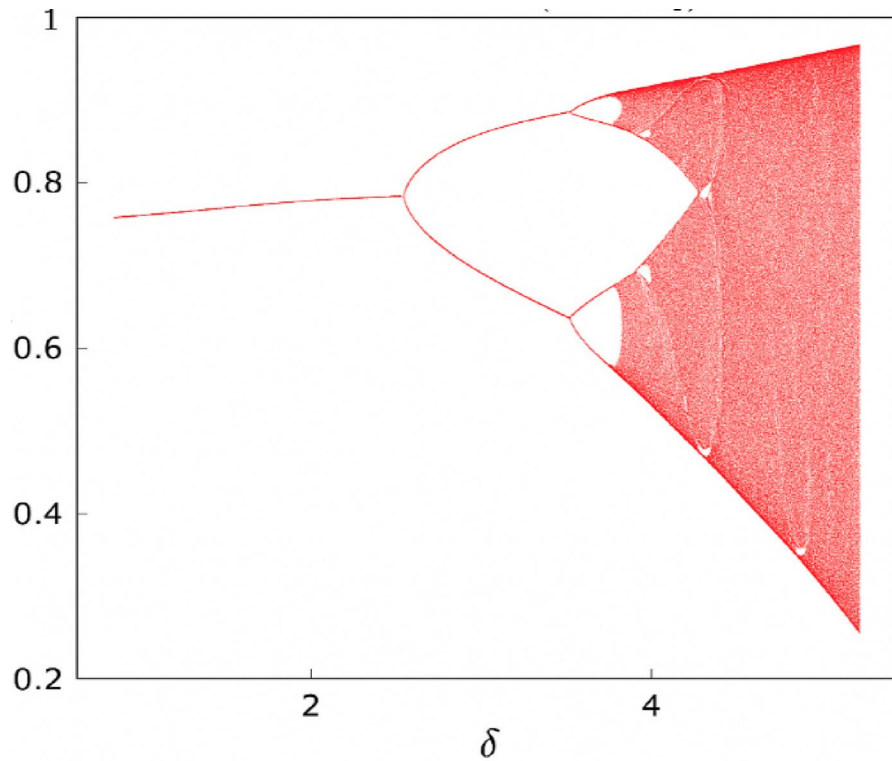


Fig. 1 Bifurcation Diagram of the Logistic Map

$$s_n = 1 - \rho s_{n-1}^2 + t_{n-1} \quad (2)$$

$$t_n = \tau s_{n-1} \quad (3)$$

where t denotes the discrete iteration index or time step in the Henon map sequence utilizing standard values $\rho = 1.4$ and $\tau = 0.3$. Figure 2 illustrates the bifurcation diagram of the Hénon map. It illustrates the variation of the state.

variable in response to changes in the control parameter . The graphic illustrates that the system initially exhibits stable periodic behavior. As ρ increases, it undergoes a sequence of bifurcations that render the dynamics chaotic. The diagram distinctly illustrates the transition from order to chaos. The densely clustered branches indicate the presence of numerous attractors, suggesting that initial conditions have a significant influence on the outcome. This behavior suggests that the Hénon map is a suitable chaotic function for cryptography, where unpredictability and sensitivity to initial conditions are crucial for secure encryption. This research uses the Henon map in conjunction with the logistic map to generate a secondary chaotic sequence. The application of bitwise XOR operations to merge the two streams produces a complex, non-linear key stream, hence enhancing the diffusion and confusion characteristics of the encryption technique. The amalgamation of chaotic systems in a hybrid configuration produces resilience against statistical and differential attacks.

To reinforce the cryptographic basis of this hybrid methodology, it is necessary to formally identify the chaotic parameters as secret keys. Because both the Logistic map

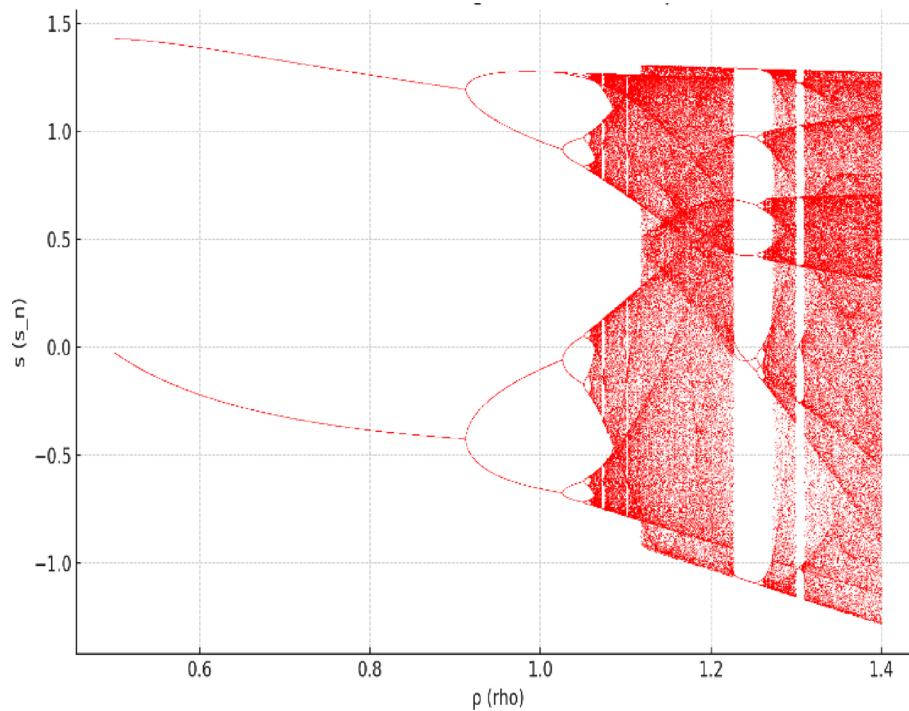


Fig. 2 Bifurcation Diagram of Henon Map

and Henon map produce extremely sensitive chaotic trajectories, the proposed system's security is determined largely by how the initial conditions and control parameters are chosen. In this context, the Logistic map is represented in Eq. (1) by δ , and y_{n-1} which can be interpreted as two independent key parts. Likewise, the Henon map expressed in Eqs. (2) and (3) employs τ , s_{n-1} , t_{n-1} , and ρ as four additional key parameters. Collectively, these create the complete initial key vector $K = (\tau, s_{n-1}, t_{n-1}, \rho, \delta, y_{n-1})$ that directs the sequence creation process. Due to the extreme sensitivity of chaotic maps to their initial values, a minuscule change to any component of K will produce an entirely distinct key stream, thereby providing a superior degree of unpredictability and cryptographic strength in the encryption process.

3.4 Zig-Zag scanning

A traversal technique employed for converting a 2D matrix into a 1D array is zigzag scanning of the matrix. Image processing effectively utilizes it to maintain the spatial relationships among pixels. This technique generates a diffusion layer from image matrix by zigzag scanning prior to encryption. The one-dimensional array produced by this scan is more effectively managed by the chaotic key stream, hence diminishing information leakage and enhancing ciphertext ambiguity. The Zig-Zag scanning of the 4×4 block matrix is shown in Fig. 3.

3.5 SHA-512

SHA-512 belongs to the SHA-2 family of cryptographic hash functions. Minor input alterations yield a uniform 512-bit output hash from data of varying lengths, hence

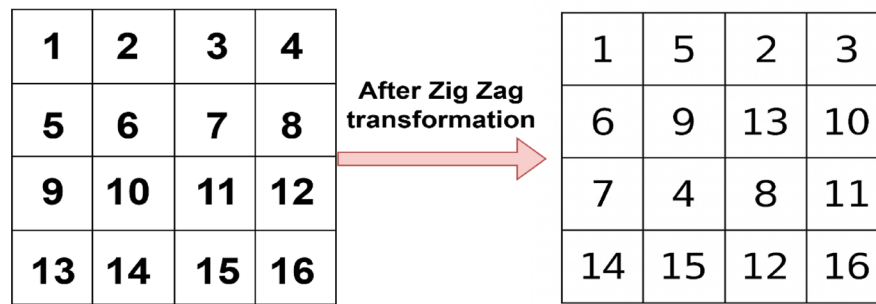


Fig. 3 Zig-Zag Scanning

guaranteeing data integrity. The proposed architecture utilizes SHA-512 to generate hash values for encrypted images, which are then transmitted along with the data to the cloud. Upon reconstruction, the hash verifies image integrity upon reception, assuring the data's immutability during transport or storage. This approach provides an enhanced integrity layer without the computational costs associated with conventional authentication techniques.

4 Methodology

This section outlines the systematic methodology for the proposed secure photo categorization and encryption framework, which utilizes edge computing, convolutional neural networks, hybrid chaotic encryption, and SHA-512 for integrity preservation in government-focused IoT systems.

4.1 System workflow

The suggested architecture is characterized by four fundamental components: IoT devices, an edge layer, cloud storage, and an optional data relay to other governmental entities. Government-utilized IoT equipment, such as document scanners or intelligent surveillance systems, gathers visual data, which is ostensibly transmitted securely to edge devices. Edge-integrated CNN models provide real-time classification of both sensitive and non-sensitive input images. The classification delineates multiple tiers of encryption to be utilized. Non-sensitive images bypass encryption, thereby avoiding redundant cryptographic processing and significantly reducing CPU utilization and transmission latency, which improves throughput for low-risk data streams. Meanwhile, sensitive images are encrypted using a hybrid chaotic approach. SHA-512 hashes aid both groups in preserving integrity. The cloud server will verify the hash after receiving the encrypted image file and its SHA-512 hash from the user. The server will calculate the hash of the image locally and compare it to the hash provided by the user. If there is a difference between the two hashes, the server will raise an integrity alert and either require a new copy of the image file to be sent or inspect it manually. The technique safeguards privacy and data security while reducing the computational load on resource-limited IoT devices. Commencing with secure IoT image acquisition, Fig. 4 illustrates the progression of the entire proposed model, which includes edge CNN-based sensitivity

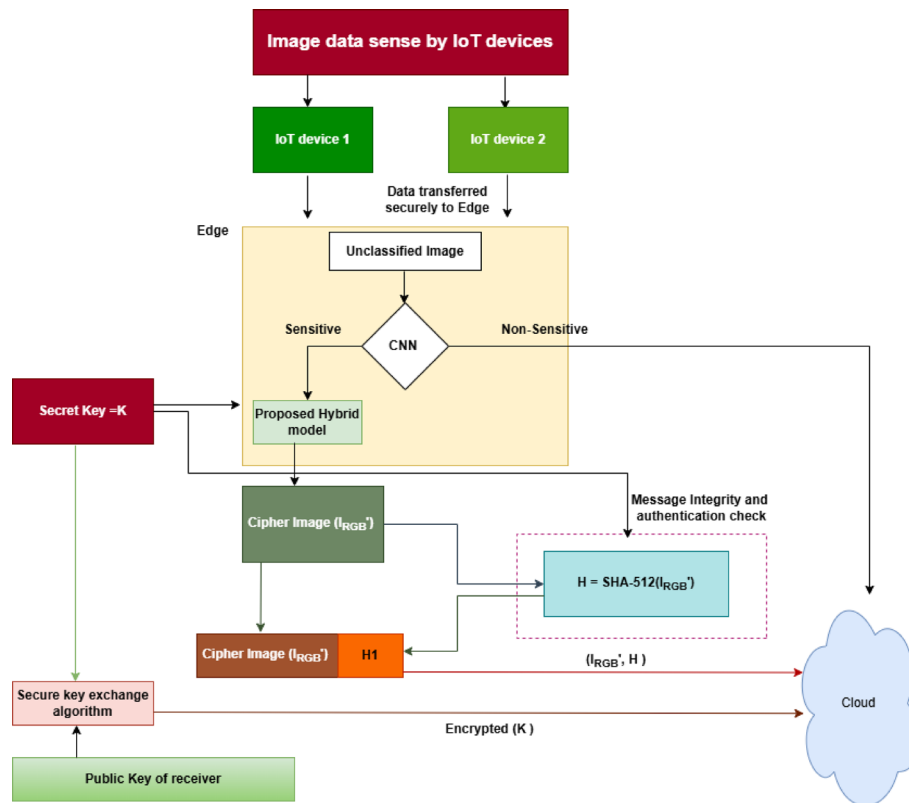


Fig. 4 Workflow of the Model

categorization, a hybrid chaotic approach for selective encryption, SHA-512 for integrity verification, and ultimate transmission to the cloud for secure government-level access.

4.2 Assumptions

1. Although this research does not concentrate on the communication between IoT devices and edge nodes, TLS/DTLS 1.3 was selected because it provides built-in resistance to downgrade attacks, supports encrypted client authentication, reduces handshake round trips, and aligns with current Government of India and global cybersecurity standards.
2. The edge device is reliable and capable of executing CNN models and hybrid encryption techniques.
3. Authorization and authentication are managed externally and are not encompassed within the recommended solution.
4. A secure and authenticated key establishment mechanism is assumed to be available prior to encryption.
5. Content-aware encryption prioritizes the integrity and confidentiality of image data.

4.3 Edge computing

The proposed system utilizes more robust edge nodes to offload resource-intensive functions from low-capacity IoT devices. This architectural decision has substantial

advantages. IoT devices are vital for collecting extensive visual data, although they generally lack the CPU power required to execute advanced deep learning models such as CNNs. Technology facilitates the categorization of photographs at the edge by guaranteeing secure data transmission to edge devices. Secondly, this offloading diminishes data transmission to the cloud and significantly alleviates the computational burden on IoT devices, hence enhancing network capacity and scalability. This system comprises three layers surrounding its edge computing unit. The Physical Layer collects data through connecting with IoT devices such as cameras, biometric scanners, and image sensors. The Network Layer and the Application Layer together comprise a CNN-based image classification engine, a hybrid chaotic encryption module, and an SHA-512-based integrity verification system facilitating secure data transfer across IoT devices, edge nodes, and cloud infrastructure. The modular, tiered design guarantees real-time processing, ensuring a secure and lightweight connection suitable for restricted government networks. This edge-enabled architecture diminishes government infrastructure costs through latency reduction, safeguards the integrity and security of private data, and enhances system efficiency.

4.4 CNN classification module

The model begins with an input layer designed to ingest $256 \times 256 \times 3$ RGB images. Following the low-to-high-level spatial characteristics of the image come three convolutional blocks, each containing convolutional filters of sizes of 32, 64, and 128, respectively. These slides go across the image to identify structural patterns, texture, and form. Every convolutional block is followed by a max-pooling layer that reduces spatial resolution while preserving the most important features. This offers translational invariance, different from computational cost-cutting strategies. Following that, a fully connected (dense) layer of 256 neurons absorb them and learns global patterns from the obtained data, therefore acting as a high-level decision-making component. Binary classification makes use of a sigmoid-activated output layer at last. A probability near 1 denotes a sensitive image, while a non-sensitive image is characterized by a probability around 0. This output specifies the type of encryption applied before transferring the image to the cloud. The model is generally light but strong enough to give quick and

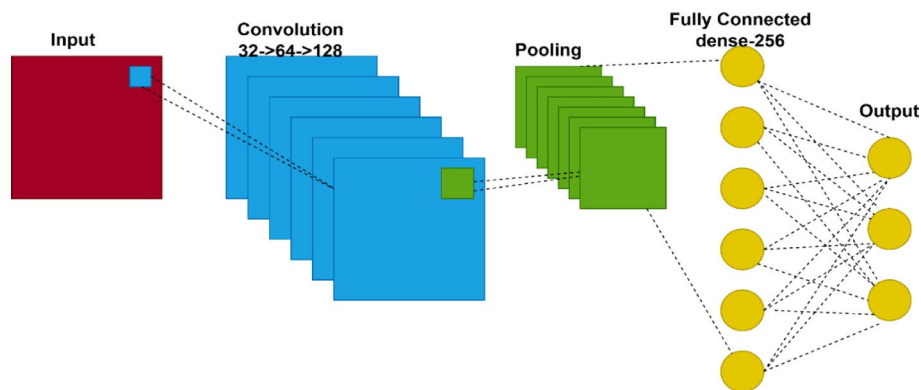


Fig. 5 Architecture of the CNN model

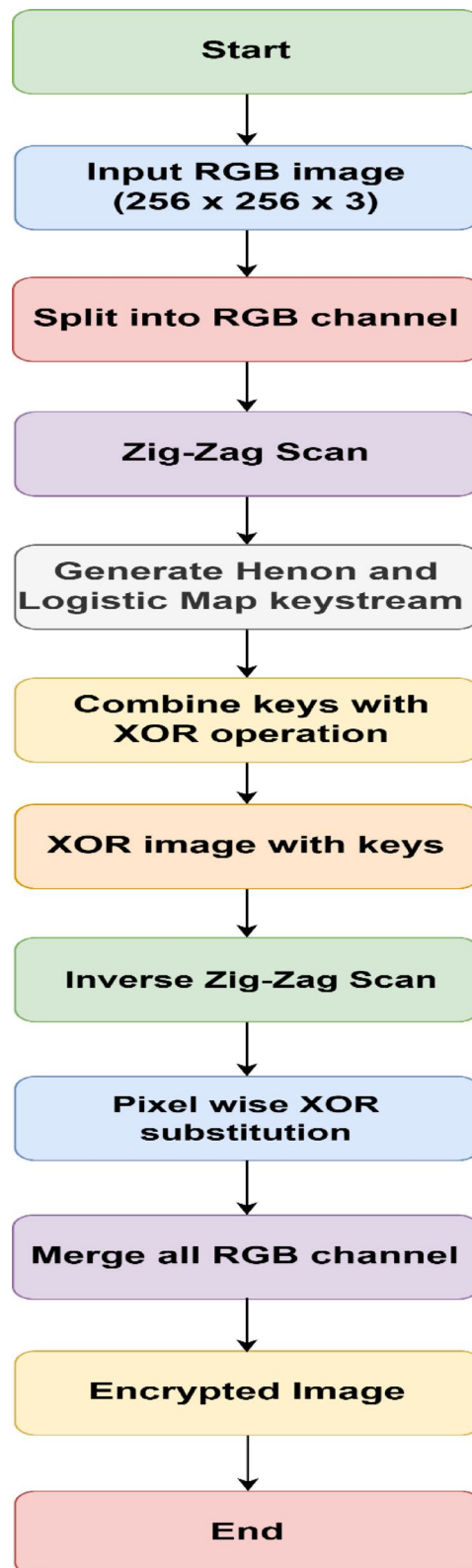
accurate sensitivity categorization just at the edge. Figure 5 illustrates the internal CNN architecture deployed at the edge node for classifying received image inputs, whether sensitive or non-sensitive. This component of the study effort is auxiliary and holds lesser significance compared to the other components. Consequently, the objective of the module was not to optimize the performance of the CNN model, but to enable the identification of functional sensitivity. This research did not do comprehensive testing of CNNs, including thorough validation, comparison of different architectures, or the use of numerous models for benchmarking. The team plans to assess more sophisticated network topologies and utilize more comprehensive evaluation approaches for future research on this project feature. Real-time edge deployment in IoT-based governmental systems is suitable for the design, as it aims to optimize accuracy, computational efficiency, and resource utilization.

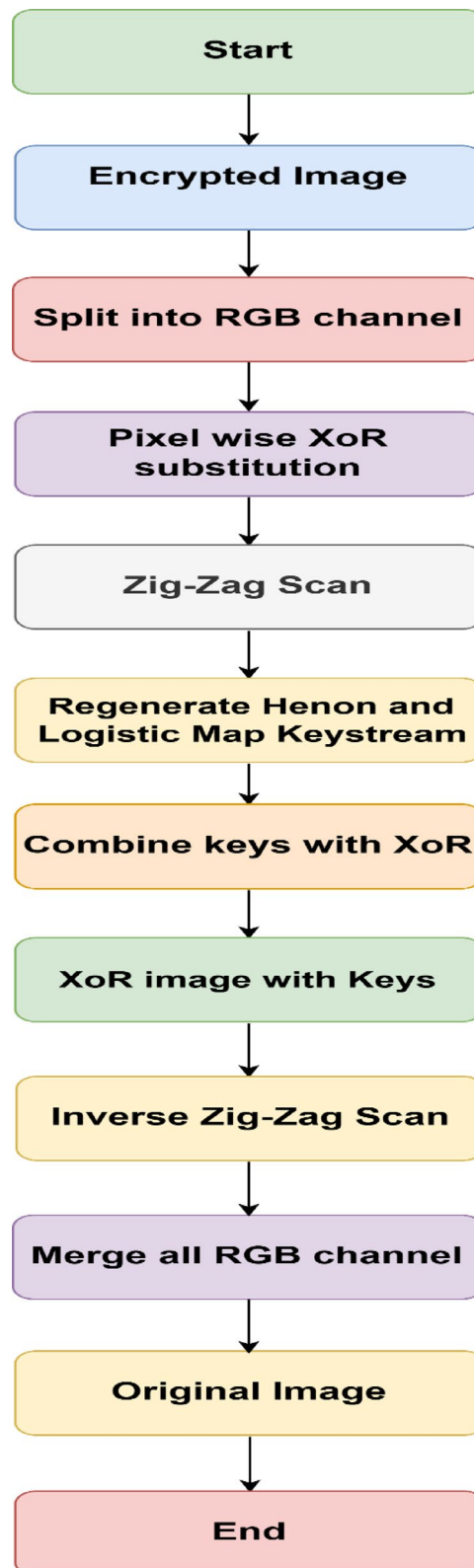
4.5 Hybrid encryption and integrity module

Depending on the CNN's classification, the sensitive images are encrypted by the proposed hybrid encryption algorithm, which begins by converting the image $I \in \mathbb{Z}^{3 \times M \times M}$ into a 1D array $P \in \mathbb{Z}^N$ of length $N = 3 \times M \times M$ using a zig-zag scan. A chaotic key stream $K \in \mathbb{Z}^N$ is generated by combining values from the Henon map and the Logistic map via bitwise XOR, such that $K_i = H_i \oplus L_i \bmod 256$, where H_i and L_i are the i -th outputs of the Henon and Logistic sequences, respectively. A feedback-based diffusion is applied, where each cipher pixel is computed as $C_0 = P_0 \oplus K_0$ and $C_i = P_i \oplus (K_i \oplus C_{i-1}) \bmod 256$ for $i = 1$ to $N - 1$ ensuring high confusion. The resulting vector $C \in \mathbb{Z}^N$ is reshaped into a 2D matrix $\tilde{C} \in \mathbb{Z}^{M \times M}$ using inverse zig-zag ordering. Finally, a pixel-wise substitution is applied where each pixel at position (i, j) is modified as $C'[i, j] = \tilde{C}[i, j] \oplus (i \cdot j \bmod 256)$, resulting in the final encrypted image C' after that, SHA-512 can be applied on the encrypted image to produce a hash for data integrity verification. The encryption and decryption flowchart of the proposed encryption model is illustrated in Figs. 6 and 7. While the non-sensitive images are sent unencrypted to the cloud.

5 Proposed model

Three basic methods are employed in the proposed model to conduct the research. The CNN model is used at the edge computing level to categorize images as sensitive or insensitive. Sensitive images are encrypted using the proposed hybrid encryption algorithms. On the other hand, non-sensitive images are sent unencrypted to the cloud. This targeted strategy optimally utilizes resources, minimizes infrastructure expenses, enhances system efficiency, safeguards confidentiality when necessary, and hence upholds integrity. Practical government applications such as digital governance, surveillance, and citizen identity management greatly benefit from this paradigm, yielding improved data privacy, expedited response times, decreased operational costs, and scalability across various public service sectors.

**Fig. 6** Encryption Flowchart

**Fig. 7** Decryption Flowchart


```

Step 1: IoT Image Acquisition
1.1 Capture image  $I$  via IoT sensor/device
1.2 Securely transmit  $I$  to edge node

Step 2: Edge-Based CNN Classification
2.1 Preprocess image  $I$  (resize to  $256 \times 256 \times 3$ , normalize)
2.2 Input  $I$  to trained CNN model
2.3 Get label  $L \leftarrow \text{CNN}(I)$ 
    If  $L = 1 \rightarrow$  Sensitive
    Else  $L = 0 \rightarrow$  Non-sensitive

Step 3: Encryption Decision Based on Sensitivity
3.1 If  $L = 1$ :
    a. Apply Zigzag scan:  $Z \leftarrow \text{Zigzag}(I)$ 
    b. Generate chaotic streams:
         $K_1 \leftarrow \text{LogisticMap}(\text{len}(Z))$ 
         $K_2 \leftarrow \text{HenonMap}(\text{len}(Z))$ 
    c. Combine streams:  $K \leftarrow K_1 \oplus K_2$ 
    d. Encrypt:  $E \leftarrow Z \oplus K$ 
    e. Apply substitution:  $E' \leftarrow \text{ModPosSub}(E)$ 
3.2 Else:
    Set  $E' \leftarrow I$  // no encryption

Step 4: Integrity Verification
4.1 Compute SHA-512 hash:  $H \leftarrow \text{SHA512}(E')$ 

Step 5: Cloud Upload
5.1 Transmit  $\{E', H\}$  to cloud
5.2 Forward securely to other government offices if needed

End Algorithm

```

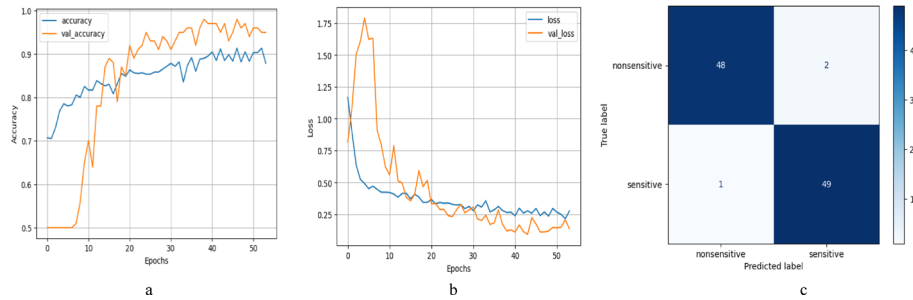
Algorithm 1 The pseudo-code for the overall architecture is presented

6 Results and discussion

A uniform computing environment was used for all tests, comprising an Intel Core processor, 16 GB of RAM, and a 256 GB solid-state drive (SSD). The Python libraries used to process the data and build the CNN model on the datasets were NumPy, Pandas, and Keras. The dataset consisted of 700 images, of which 350 were considered sensitive, and 350 were non-sensitive. Six hundred images were used for model development, with an internal 80/20 split applied, resulting in 480 training images and 120 validation images. The remaining 100 images were used for testing, with 50 images from each class. Five examples of sensitive images have been selected to illustrate the encryption capabilities of the combined chaotic scheme, which were converted to a $3 \times 256 \times 256$ resolution (Fig. 8). These examples were chosen based on their importance in government IoT data processing related to the following areas of highest risk documents: Aadhaar cards, PAN cards, driving licenses, bank passbooks, and voter identification cards. The standard Baboon image, with a resolution of 256×256 pixels, was also used as the basis of comparison, as it remains the benchmark for chaotic encryption literature. A comparison of results using Baboon was performed in the experiment to confirm that the approaches had the same level of performance as those in all other studies in the chaotic encryption literature.

6.1 CNN classification performance

Favorable assessment results obtained during testing further enhance the effectiveness of the proposed CNN model in classification. The model effectively extracted distinguishing features to separate sensitive from non-sensitive photos, achieving training accuracy of approximately 98.6%, validation accuracy of around 97.2%, and testing accuracy of

**Fig. 8** Sensitive Images**Fig. 9** **a** Training and validation accuracy curves, **b** training and validation loss curves, and **c** confusion matrix evaluated on the test dataset, showing testing accuracy of 96.8%, training accuracy of 98.6%, and validation accuracy of 97.2%

96.8%. Furthermore, the model achieves an Area Under the Curve (AUC) value of 0.98, indicating its exceptional ability to accurately differentiate between the two groups and its high precision in binary classification. Figure 9(a) illustrates the consistent convergence of accuracy curves for both training and validation throughout the epochs, while and 9(b) demonstrates a continual decline in loss without evident overfitting. Figure 9(c) presents the confusion matrix, which provides insights into the accuracy of the categorization. The model accurately identified 49 out of 50 sensitive photographs and 48 out of 50 non-sensitive photos from a total of 100 test photos. This demonstrates its exceptional ability to distinguish between the two types of images. The model exhibits high sensitivity and specificity, evidenced by an Area Under the Curve (AUC) value of 0.98. This renders it ideal for real-time edge-based sensitivity classification. The CNN module initiates the process of selective encryption. This study does not primarily concentrate on it, despite its efficacy.

6.2 Proposed encryption algorithm performance

Encryption performance was evaluated based on Time complexity, encryption/decryption time, throughput, statistical attacks, and Differential attacks.

6.2.1 End-to-end latency

Latency refers to the duration required for an image to navigate through the IoT-edge-cloud pipeline, encompassing the processes of capturing, sorting, encrypting, hashing, and transmitting the image. In secure, real-time IoT contexts, precisely measuring end-to-end latency is crucial for evaluating system responsiveness and its suitability in real-world scenarios. Using the Baboon image of size $3 \times 256 \times 256$ (27.2 KB) as a representative benchmark, the end-to-end latency of the proposed framework can be modeled as $T_{\text{total}} = T_{\text{acq}} + T_{\text{tx,IE}} + T_{\text{pre}} + T_{\text{cnn}} + T_{\text{enc}} + T_{\text{hash}} + T_{\text{tx,EC}} + T_{\text{ver}} + T_{\text{cloud}}$. For this image, the acquisition time T_{acq} is approximately 20 ms, the preprocessing time T_{pre} is around 10 ms, and the CNN inference time T_{cnn} ranges from 90 to 100 ms. The most time-consuming step, the hybrid chaotic encryption step, requires $T_{\text{enc}} = 1.21$ s. The hashing time T_{hash} is generated in around 0.16 s. Data transmission time $T_{\text{tx,IE}}$ From the IoT to the edge takes 20 milliseconds, and data transmission $T_{\text{tx,EC}}$ from the edge to the cloud requires around 40 milliseconds. The cloud verifies the hash in T_{ver} time is around 0.11 s, and the data is stored on the cloud at T_{cloud} is 10 milliseconds. The entire end-to-end latency is around $T_{\text{total}} = 1.67$ to 1.72 s. This indicates that encryption is the primary cause of the delay, whereas CNN inference and network transfer contribute minimally to the overall duration. This latency profile demonstrates that the system can effectively manage $3 \times 256 \times 256$ images in nearly real-time, which is commendable for secure government IoT implementations.

6.2.2 Time complexity

A solitary traversal of the pixel array is achieved by synthesizing an inverse chaotic key stream, utilizing Henon and logistic maps, feedback-driven XOR diffusion, modular position-based substitution, and zigzag scanning, resulting in linear computational expense. The suggested optical encryption method exhibits a linear time complexity of $O(N)$. The SHA-512 hash is calculated sequentially for integrity verification, although it is optional. Our solution ensures that the entire encryption process maintains a temporal complexity of $O(N)$. The comparison of time complexity with various state-of-the-art algorithms is presented in Table 1, revealing that the proposed algorithms exhibit linear complexity, making them computationally efficient and suitable for real-time applications in resource-constrained environments, such as fog computing nodes.

6.2.3 Encryption/Decryption time

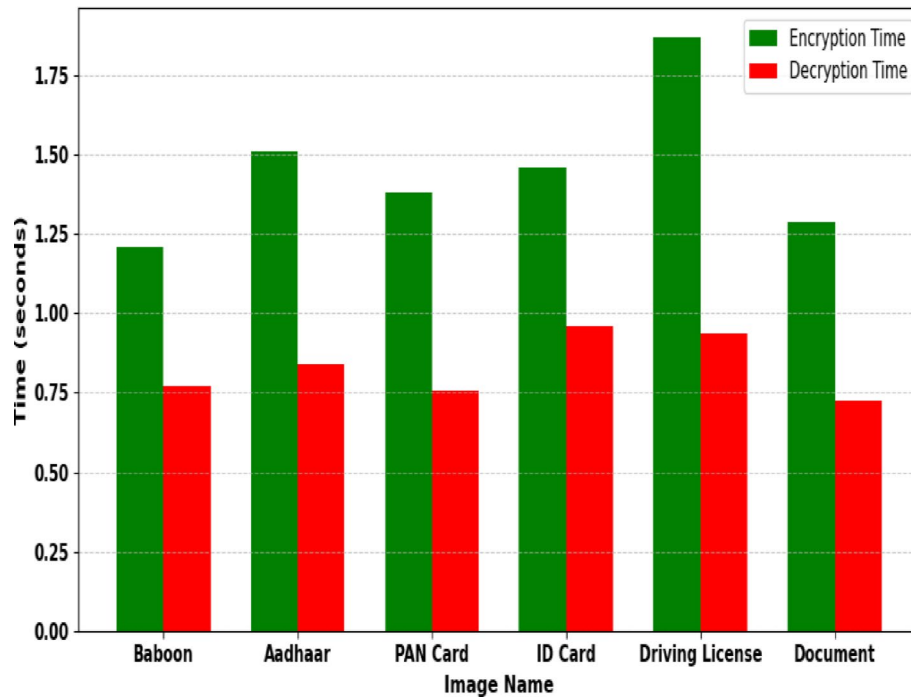
Two components of the proposed framework govern the encryption time: T_{hash} is the time defined by the SHA-512 hashing algorithm for integrity verification, whereas T_{enc} represents the time utilized by the hybrid chaos-based encryption method employing Logistic and Henon maps. The sum of T_{enc} and T_{hash} thus governs the entire encryption duration. Five sensitive images were selected for security assessment: an Aadhaar

Table 1 Time complexity analysis

Algorithm	Time complexity
Qiang et al. [37]	$O(N^2)$
Alrubaie et al. [38]	$O(N)$
Gabr et al. [39]	$O(N^2)$
Proposed Work	$O(N)$

Table 2 Encryption/Decryption time and throughput

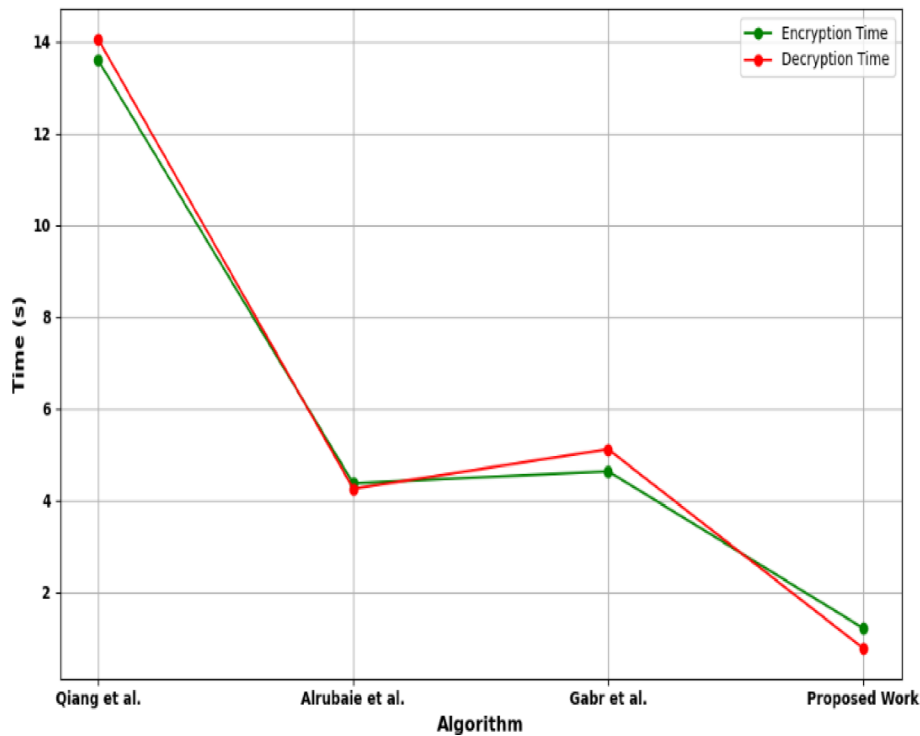
Image	Size (Kb)	Encryption time (s)	Decryption time (s)	Encryption throughput (Kb/s)	Decryption throughput(Kb/s)
Baboon	27.2	1.21	0.772	22.48	35.23
Aadhar	70.8	1.51	0.841	46.89	84.20
Pan Card	127	1.38	0.759	92.03	167.27
ID Card	52.3	1.46	0.960	35.82	54.48
Driving License	124	1.87	0.935	66.31	132.62
Document	38.9	1.29	0.724	30.16	53.74

**Fig. 10** Encryption/decryption time (sec)

card, a PAN card, a bank passbook, a driving license, and an ID card. The encryption times for these images are shown in Table 2, and a visual representation is also provided in Fig. 10. In addition to encryption, the decryption duration was calculated to assess the viability of secure real-time communication and is presented in Table 2, along with a visual representation in Fig. 10. This approach reveals the decryption duration using SHA-512, specifically the time required to reverse the encryption process and verify the integrity of the obtained image. It encompasses the duration necessary to recalculate and compare the hash T_{ver} alongside the time needed to decode the cipher image T_{decr} utilizing a hybrid chaos-based methodology. The total decryption time is therefore $T_{\text{ver}} + T_{\text{decr}}$. To ensure methodological consistency, the existing models reported in [33–35] were evaluated using the standard Baboon image, which is widely used in image encryption research. For fairness and direct comparability, the same benchmark image was also used in the proposed implementation, enabling a uniform basis for assessing encryption time, decryption time, throughput, and statistical metrics across all methods. Table 3 presents a comparative analysis of the encryption/decryption times of the proposed work with existing algorithms for the Baboon image for better clarification. Figure 11 clearly demonstrates that the proposed approach, averaged across three

Table 3 Comparison of the proposed work with other algorithms

Algorithm	Encryption time (s)	Decryption time (s)	Encryption throughput (Kb/s)	Decryption throughput(Kb/s)
Qiang et al. [37]	13.59	14.05	2.00	1.93
Alrubaie et al. [38]	4.37	4.25	6.22	6.40
Gabr et al. [39]	4.630	5.11	5.87	5.32
Proposed Work	1.21	0.772	22.48	35.23

**Fig. 11** Comparison of Time(sec)

representative baseline schemes, exhibits a substantial computational advantage, achieving a 6.22-fold reduction in encryption time and a 10.11-fold reduction in decryption time.

6.2.4 Encryption/Decryption throughput

Encryption and decryption throughput refers to the speed at which images are converted from plaintext to ciphertext and vice versa. This parameter is essential for assessing the performance of real-time systems. The encryption/decryption throughput is given by Eq. 5.

$$\text{Throughput} = \frac{\text{Size of Data (Kb)}}{\text{Encryption or Decryption time in sec}} \quad (4)$$

The throughput value for each sensitive image is given in Table 2, and a bar graph is shown in Fig. 12. Upon assessing five real-world sensitive image samples, the proposed method demonstrated significantly superior throughput compared to conventional encryption techniques, as shown in Table 3 and illustrated in Fig. 13. On average, the proposed approach delivers a 6.23× increase in encryption throughput and a 10.12×

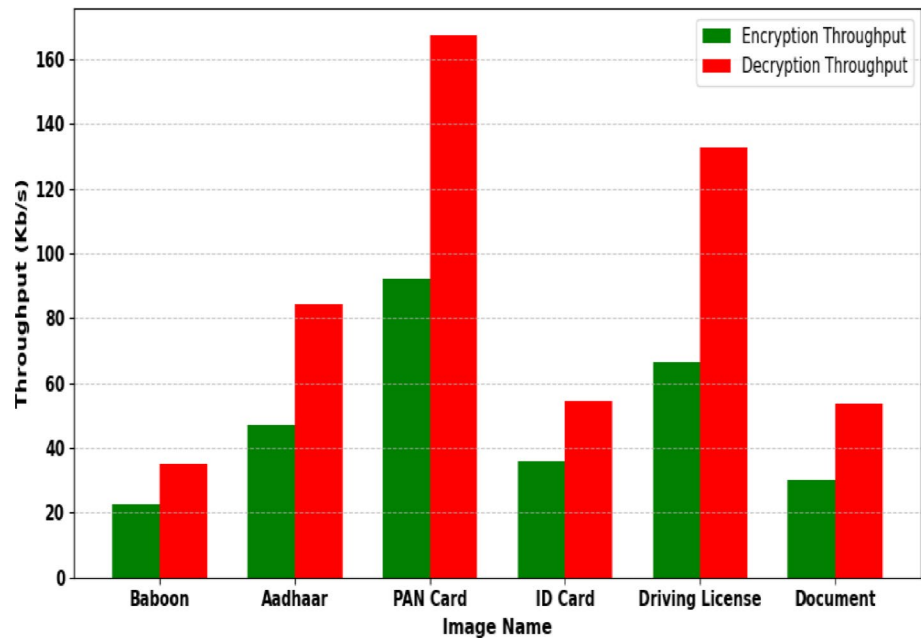


Fig. 12 Encryption/decryption throughput (kb/s)

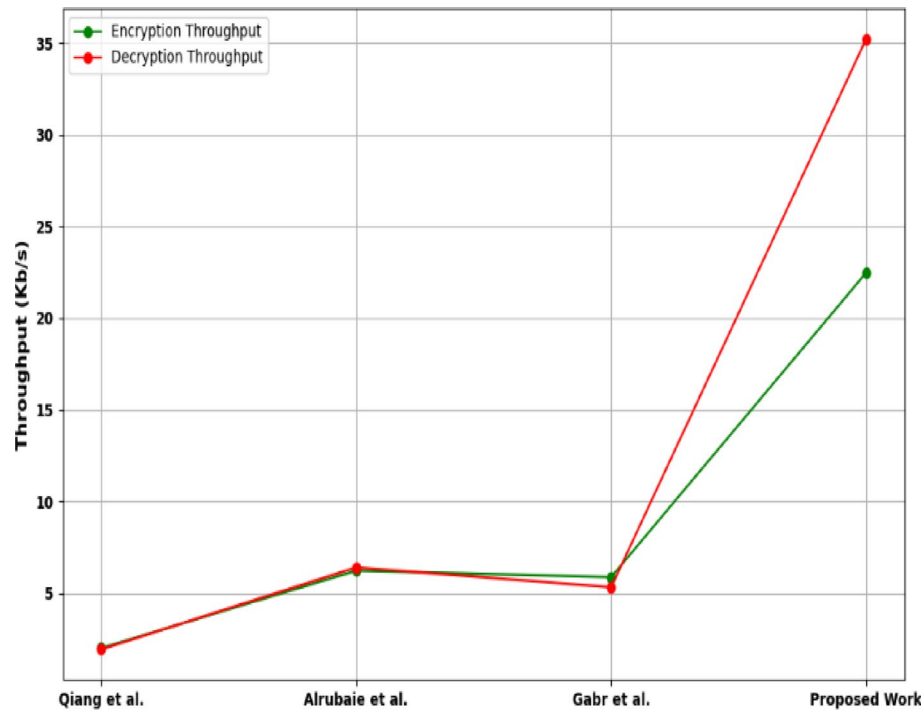


Fig. 13 Comparison of throughput (Kb/sec)

increase in decryption throughput relative to representative baseline schemes. This improvement is primarily attributed to the lightweight cryptographic operations and optimized processing flow employed in the proposed design.

6.2.5 Differential attack analysis

An evaluation of differential attack analysis provides a measure of how well a given image encryption method can defend against an adversary exploiting the correlation between small variations in a plaintext image and their resultant effect on the encrypted image. For the proposed hybrid chaotic encryption method, strong resistance against differential or selective-plaintext attacks has been demonstrated through the use of small, controlled changes in a sensitive plaintext image to generate substantially different encrypted outputs. Strong diffusion and confusion characteristics are required by encryption methodologies used to perform selective encryption since they are focused on protecting specific parts of the image. Thus, if it is possible for an attacker to determine the structure of the plaintext image based upon the difference between variations of the ciphertext; then, the system being examined will not exhibit the qualities necessary to resist this type of attack. To determine the degree of differential resistance of the proposed system, two commonly used metric, the Number of Pixel Change Rate (NPCR) and the Unified Average Change Intensity (UACI), were used. These two metrics provide the means of measuring the extent to which a changed ciphertext image differs due to one pixel being changed in the corresponding plaintext image. Let $I_1(i, j)$ and $I_2(i, j)$ denote the two encrypted images obtained from plain texts that differ in only one pixel. The NPCR and UACI are mathematically computed as given by the equations 5 and 6 respectively.

$$NPCR = \frac{\sum_{i=1}^M \sum_{j=1}^N C_{ij}}{M \times N} \times 100 \quad (5)$$

$$\text{Where } C_{ij} = \begin{cases} 0, & \text{if: } I_1(i, j) = I_2(i, j) \\ 1, & \text{if: } I_1(i, j) \neq I_2(i, j) \end{cases}$$

$$UACI = \frac{\sum_{i=1}^M \sum_{j=1}^N |I_1(i, j) - I_2(i, j)|}{M \times N \times 255} \times 100 \quad (6)$$

A pixel in all sensitive document images was altered, and all versions of both plaintext documents were encrypted using the same key vector K. Table 4 summarizes the NPCR and UACI values of all the encrypted document outputs. The proposed hybrid chaotic model produces NPCR values greater than 99.55% and UACI values of approximately 33%, matching the theoretically predicted NPCR and UACI values of 99.60% and 33.46%, respectively, for an ideal 8-bit chaotic encrypted system as outlined by Midoun in [40]. The results confirm that the proposed model has a high level of diffusion, making the exploitation of plaintext/ciphertext relationships by attackers nearly impossible. Table 5 contains a comparative analysis of the NPCR and UACI values produced by the proposed method and the currently available techniques. The proposed method exhibits

Table 4 NPCR and UACI value for different images

Image Name	NPCR (%)	UACI (%)
Aadhaar	99.59	33.42
PAN Card	99.62	33.37
Driving License	99.55	33.28
ID Card	99.57	33.31
Bank Passbook	99.61	33.46
Baboon	99.64	33.44

Table 5 Comparison of NPCR and UACI with other algorithms for the image baboon

Algorithm	NPCR	UACI
Proposed	99.64	33.44
Qiang et al. [37]	99.61	--
Alrubaie et al. [38]	99.55	32.81
Gabr et al. [39]	99.65	30.4567

NPCR and UACI values that are consistently greater than or equal to those produced by currently available techniques.

6.2.6 Statistical attack analysis

The security evaluation of the proposed hybrid chaos-based encryption method reveals its considerable robustness against statistical attacks. These attacks are typically utilized to discern patterns in encrypted data that could aid in the retrieval of the original material. Histogram uniformity confirms pixel-value randomization; Entropy nearing 8 indicates maximal unpredictability; and correlation coefficients ≈ 0 validate loss of spatial relationships between adjacent pixels. Collectively, these demonstrate high diffusion and confusion strength, essential for cryptanalytic resilience.

6.2.7 Histogram analysis

To evaluate the accuracy of encryption, histogram analysis, which offers details about the distribution of pixel values, is an essential technique in image processing and analysis. The proposed approach has produced encrypted images with a uniform distribution of pixel values, indicating effective encryption, according to the histogram analysis of the six sensitive images such as Aadhar Card, PAN, ID card, Driving License, and Document shown in Fig. 14(a), (b), (c), (d), and (e) respectively. The lack of apparent patterns or structures within the histograms validates the suggested encryption technique's dependability, guaranteeing the secure masking of sensitive image data. Governments can utilize the proposed model to facilitate secure communication between IoT devices, edge devices, and the cloud.

6.2.8 Correlation coefficient

One way to quantify an image's unpredictability is through correlation analysis. In an ideal random-like image, the correlation between adjacent pixels is low to prevent information from escaping [41]. The CC between two information sequences can be calculated mathematically by Eq. 7.

$$\text{corcoef}(p, c) = \frac{E[(p - \mu_p)(c - \mu_c)]}{\sigma_p \sigma_c} \quad (7)$$

where c represents the adjacent pixels corresponding to p in the diagonal, horizontal, and vertical directions. The means of p and c are denoted by μ_p and μ_c . The standard deviation of p and c is represented by σ_p and σ_c . $E[\cdot]$ is the expected value. The absolute values of the coefficients range from 0 to 1. There is no linear relationship between the plain and cipher images when the correlation between the two sequences is low, as indicated by an absolute correlation value close to zero. The model employed correlation coefficient analysis on the red, green, and blue channels to examine the encrypted versions of

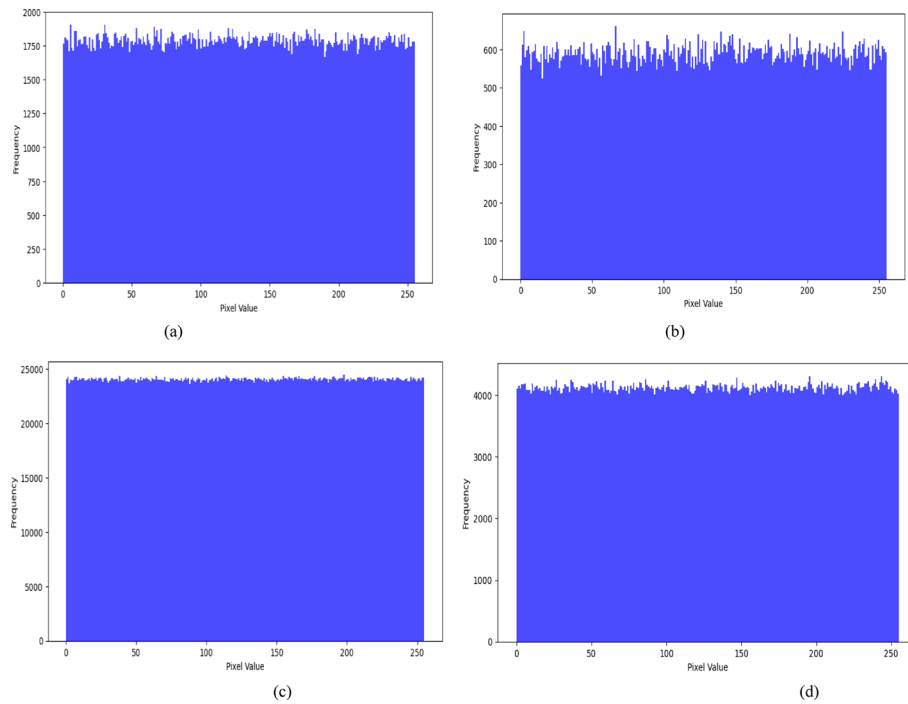


Fig. 14 a Adhar Card b PAN. c ID Card d Driving license. e Documents

color photographs. The correlation coefficients for all images across all orientations and color sections are about zero, as indicated in Table 6. The Baboon image had red-channel correlation values of -0.0015 (horizontal), 0.0004 (vertical), and -0.0005 (diagonal). The other images also exhibited consistent patterns, indicating that the encryption functioned uniformly on each occasion. The correlation scatter plots for the image Baboon are shown in Fig. 15. This data further substantiates the claim by illustrating a random and unstructured distribution of pixel intensity pairings, which plainly indicates that spatial redundancy is disrupted. The tenuous connection between encrypted baboon and Aadhaar images indicates that the proposed encryption technique is sufficiently robust to thwart statistical attacks and complicate predictive analysis. Furthermore, the model is compared with the average correlation coefficients with other state-of-the-art algorithms, as shown in Table 7, and found that the proposed model has the lowest CC among all.

6.2.9 Entropy analysis

The information entropy of an encrypted image serves as an essential quantitative measure for protecting algorithms from statistical attacks. This entropy reflects the statistical properties of the grayscale distribution throughout the image. When the computed entropy value nears 8, it indicates that the encrypted image is likely to appear as random noise, implying a consistent grayscale distribution [42]. Equation 8 represents the formula used to calculate information entropy.

$$E = - \sum_{t=0}^M p(t) \log_2(p(t)) \quad (8)$$

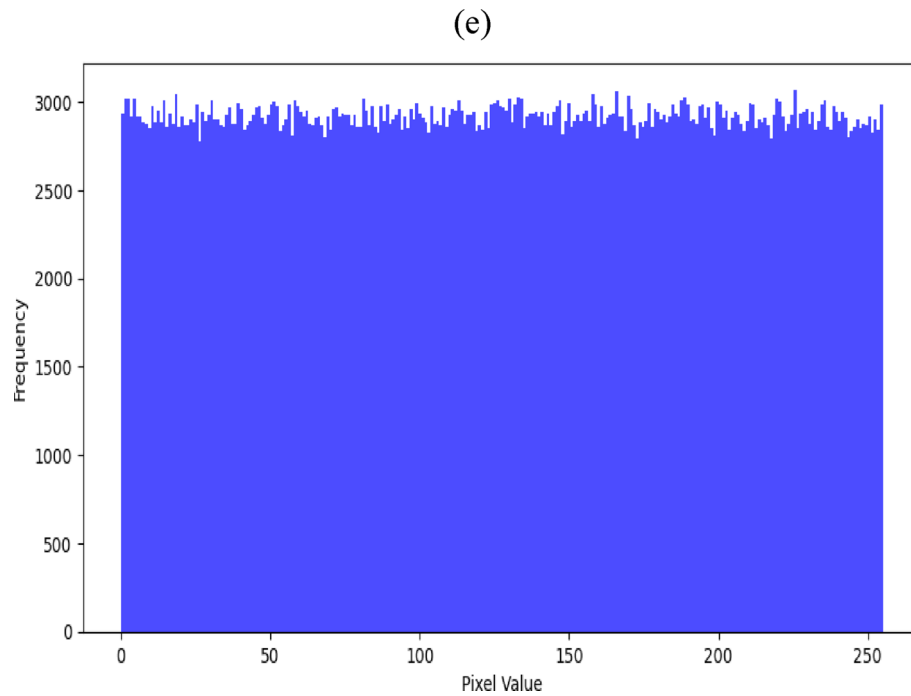


Fig. 14 (continued)

Table 6 Analysis of correlation coefficients

Image	Red Channel			Green Channel			Blue Channel		
	H	V	D	H	V	D	H	V	D
Baboon	-0.0015	0.0004	-0.0005	-0.0012	-0.0002	0.0008	-0.0013	-0.0020	-0.0004
Aadhaar	-0.0014	0.0003	-0.0004	-0.0010	-0.0001	0.0006	-0.0018	-0.0016	-0.0006
Driving License	-0.0018	-0.0048	0.0036	-0.0010	-0.0034	-0.0021	-0.0037	-0.0016	-0.0034
ID Card	-0.0018	-0.0054	-0.0015	-0.0013	-0.0071	0.0012	-0.0002	-0.0072	0.0002
PAN Card	-0.0002	-0.0038	-0.0031	-0.0013	-0.0063	0.0024	0.0001	0.0040	-0.0031
Documents	0.0014	-0.0042	-0.0036	0.0019	-0.0041	-0.0030	0.0018	-0.0039	-0.0032

In this context, $f()$ signifies the likelihood of the pixel value appearing, while M indicates the overall count of color levels associated with pixel values. The results of the entropy calculations for the encrypted image are displayed in Table 8, indicating that the proposed encryption method attains an information entropy value close to the optimal level of 8. In Table 9, the model compares the average entropy of existing methods for the image Baboon, providing an unbiased analysis. It is found that the suggested model generates the most random and unpredictable output. Thus, it is the safest alternative for image encryption at Edge computing.

Multiple analytical criteria indicate that the proposed encryption technique displays considerable resilience against statistical attacks. The information entropy scores for all encrypted color images constantly approach eight, the ideal level. This signifies that deciphering the content of the encrypted message is exceptionally challenging. The correlation coefficients of adjacent pixels across all RGB channels, irrespective of horizontal, vertical, or diagonal orientation, are almost zero. This disrupts the connections among the components of images. The histograms of the encrypted images indicate that the intensity levels are uniformly distributed. This indicates that hackers are unable

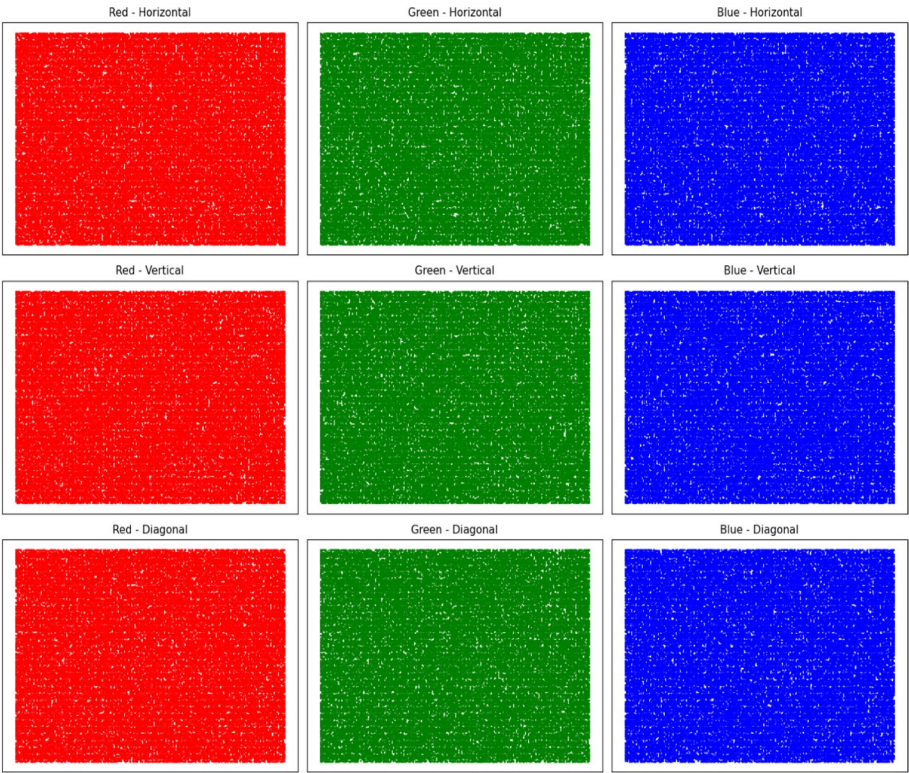


Fig. 15 Correlation plot for encrypted Baboon Images of Red, Green, and Blue channels on each dimension

Table 7 Comparison of average correlation with existing methods

Algorithm	Average CC
Qiang et al. [37]	0.0037
Alrubaie et al. [38]	0.0432
Gabr et al. [39]	0.003282
Proposed Work	0.00092

Table 8 Entropy analysis

Image	Red	Green	Blue	Average
Baboon	7.9976	7.9971	7.9969	7.9972
Aadhaar	7.9977	7.9976	7.9976	7.9976
PAN card	7.9971	7.9969	7.9972	7.9971
Driving license	7.9972	7.9968	7.9971	7.9970
ID card	7.9972	7.9969	7.9968	7.9970
Documents	7.9976	7.9974	7.9974	7.9975

Table 9 Comparison of average entropy with existing algorithm

Algorithm	Average entropy
Qiang et al. [37]	7.9974
Alrubaie et al. [38]	7.9898
Gabr et al. [39]	7.9856
Proposed Work	7.9972

to discern any patterns in the data that they can exploit. The findings indicate that the encryption technique surpasses statistical dependence and effectively safeguards against statistical attacks.

6.2.10 Key space analysis

The principal measurement of a cryptosystem's strength is the size of its key space. A large key space is necessary to ensure adequate protection from brute force attacks, as it makes it very difficult for an attacker to perform an exhaustive search on every possible key. Alvarez and Li in [43] suggested that an adequate amount of security for an image encryption algorithm requires that the key space is larger than 2^{100} , meaning that an exhaustive search will not likely yield a solution in a timely manner. As for Chaotic encryption, the secret key $K = (\tau, s_{n-1}, t_{n-1}, \rho, \delta, y_{n-1})$ has six independent parameters' each represented by a double precision floating point number (64-bit), and therefore each parameter has about 50 bits of entropy. Consequently, the combined entropy of the secret key exceeds 300 bits, resulting in an effective key space greater than 2^{300} . This value represents a conservative lower bond on the actual key space. Therefore, the proposed encryption model provides a key space that is vastly larger than the minimum recommended threshold and is computationally infeasible to exhaust using current brute-force techniques. Table 10 outlines the size of the key space for the proposed model compared with those of previously developed chaotic methods for image encryption. To recover the original image, an adversary will need to determine the values of all six independent parameters that comprise the secret key making brute force cryptanalysis virtually impossible.

6.2.11 Key sensitivity analysis

The chaos-based image encryption system must have sensitivity to the secret key where even the smallest change made to the key results in producing a new coded output. In the proposed hybrid design, the initial values and parameters used to create the secret key exhibit a large amount of sensitivity, therefore providing a strong level of protection against differential key attacks. To confirm this property, a set of standard document images was encrypted utilizing the designated secret key K and then each image was encrypted with the new key $' = K + 10^{-15}$ (modified secret key). The cipher texts produced with K and $'$ were compared and evaluated based on two criteria: NPCR and UACI. From Table 11 the results for NPCR values are higher than 99.55% and UACI results were close to the ideal values of 33% for each image. This demonstrates the exceptional diffusion and avalanche effect enabled by the hybrid structure of the chaotic encryption system.

Table 10 Comparative analysis of key space

Algorithm	Key space
Proposed	2^{300}
Qiang et al. [37]	2^{238}
Alrubaie et al. [38]	2^{160}
Gabr et al. [39]	2^{372}

Table 11 NPCR and UACI based on key sensitivity

Image Name	NPCR	UACI
Aadhaar	99.58	33.29
PAN Card	99.62	33.41
Driving License	99.55	33.18
ID Card	99.59	33.34
Bank Passbook	99.63	33.47
Baboon	99.67	33.52

7 Limitations of the study

Although the use of a CNN-assisted hybrid chaotic encryption framework to protect sensitive government document images has proven feasible as well as effective, there are still many limitations that offer opportunities for future improvements.

- The first limitation is that although the architecture has been designed to be used in edge computing environments, all testing was completed in a computer workstation with an Intel Core i7 processor, and consequently the results do not represent actual edge device performance metrics including but not limited to energy usage, CPU use, memory usage, and thermal performance when operating on limited resource environments like Raspberry Pi, Jetson Nano, Coral TPU devices or IoT microprocessors. Additionally, the latency calculated for this research project as being the latency of this proposed hybrid encryption framework using CNN, does not encompass a complete end-to-end calculation of how long it takes for an entire camera capture process, including how long it takes to perform all four portions of the hybrid encryption process: capture, image conversion into a CNN model and transform the image back to an image, encryption time for the capture, network transfer of the encrypted image to a cloud system, decryption of the encrypted image using decryption protocols developed for consuming encrypted images within heterogeneous IoT environments.
- The CNN classifier utilized to detect sensitivity is intentionally a lightweight classification model that is intended to be a functional trigger for selective encryption, rather than a primary contribution of this work. Therefore, this paper does not present the use of typical machine learning methods (such as cross-validation and multiple runs) for the CNN model or other classification models; nor does it present a statistical analysis of the model; nor does it provide established confidence intervals; nor does it present benchmark results of the architecture of CNNs when compared to other lightweight CNNs such as MobileNet, ShuffleNet, and EfficientNet-Lite. The misclassification handling is not entirely explored, including a thorough examination of the worst-case failure modes (such as false negatives) and their effects on data exposure. Furthermore, the performance evaluation described in this paper does not incorporate federated updates, client drift, and personalized approaches as they relate to government-scale IoT applications.
- Third, although the encryption model integrates a hybrid chaotic permutation–diffusion mechanism and reports comprehensive statistical and differential analyses including entropy, correlation, NPCR, UACI, and key-sensitivity behaviors but there are still limitations due to the lack of actual security against chosen-ciphertext, known-plaintext attacks, and even unknown plaintext attacks. This means that further experiments need to be conducted to validate the original claims of this

research paper. In addition, while single-run performance metrics are reported, multi-run variance analysis and formal confidence-interval calculations have not been performed, which would further strengthen the statistical robustness of the results. Furthermore, as stated previously, while there are references in this article to the secure generation and distribution of keys in a way that is conceptually defined, there are no references to actual implementations or testing of the key generation process. Finally, although this work focuses on classical models of cryptography and chaotic maps as a form of information security, it does not acknowledge that there are many post-quantum or quantum-resistant security models that should be integrated into the modern designs for protecting information due to the current ability of quantum computing to break any current data encryption methodologies. In order for governments and organizations to protect their information in the IoT environment from potential future threats of quantum computing, implementation of current post-quantum designs such as those based on lattices or hash functions should be integrated with current systems for exchanging keys and protecting digital identity verification via digital signatures.

- As the dataset that was used for training the CNN (700 images, 350 sensitive images, 350 non sensitive images) is limited both in its scale and diversity. While the dataset was a sufficient size for a proof-of-concept validation, it does not allow for evaluations of the variability of real-world images due to differences in lighting, backgrounds, camera quality, and/or ID cards that have been worn or damaged. The reason for the training time accuracy curves is due to how the internal 80/20 validation split was split into two sets. Future work will need to include the performance of large-scale stress tests, where millions of images are used, as part of the requirement to validate the deployment of this technology at the national level.
- Finally, there is still no empirical testing of the capability of the architecture to provide system-level scalability with respect to the volume of parallelized data streams coming from hundreds or thousands of edge nodes. Many areas remain unaddressed regarding synchronizing between Cloud and Edge, Load Balancing of multiple Nodes, Delays associated with message queues, Decentralized Integrity verification mechanisms for the network, and Fault Tolerance related to system outages or failures. All of the above-mentioned elements need to be considered for practical application to Smart Government, E-Government, or Public Service IoT environments.

8 Conclusion and future direction

This research provides an all-inclusive framework for handling images securely and effectively in IoT infrastructures that are primarily utilized and developed by governments. The framework combines the use of a CNN for classifying the sensitivity of images based on the analysis of image content into categories of sensitive and non-sensitive images, with a hybrid chaotic encryption-based mechanism. The framework utilizes a lightweight edge-assisted model with minimal computational overhead by encrypting only the images classified as sensitive. This approach leverages the use of Logistic–Henon chaotic sequences and hashing algorithms, such as SHA-512, to provide robust confidentiality and integrity protection for all images processed within the infrastructure. The experimental results indicate that the classification accuracy was

reasonably high (training $\approx 98.6\%$, validation $\approx 97.2\%$, testing $\approx 96.8\%$) despite that there is very low inter-pixel correlation in the encrypted images; the encrypted images have Entropy values that are very close to ideal (~ 7.997), and the NPCR/UACI performance is high, indicating robust confusion and diffusion characteristics. The hybrid encryption scheme provided a significant increase in processing throughput, approximately 6.23 times that of all other conventional approaches, making the proposed architecture suitable for deployment in real-time within resource-constrained governmental IoT environments. This development of the integrated selective-encryption approach provides a viable solution to balance accuracy, security, latency, and computational efficiency for the purposes of securing and managing images using IoT technologies that operate in conjunction with governmental entities.

The Future Work will address the remaining cryptographic evaluations, as well as the deployment-level evaluations, for which the scope of this method did not include. The additional cryptographic evaluations will include performing multi-run statistical tests, such as estimating confidence intervals, and benchmarking against chosen-plaintext and known-plaintext attack models. In addition, practical methods for key management will be developed, tested, and documented to ensure the safe generation, distribution, rotation, and revocation of keys within multi-edge IoT environments. Documents describing the deployed system running on actual edge hardware will be developed and tested to assess the limitations in terms of energy consumption, memory utilization, and edge/cloud synchronization delays. This research will assess the compatibility of the cryptographic scheme with emerging post-quantum security primitives, thus establishing long-term cryptographic resilience for extensive government IoT deployments.

Author contributions

S. Afzal contributed conceptualization of the research problem, design of the lightweight CNN architecture, development of the hybrid chaos-based encryption algorithm, and drafting of the initial manuscript. M. Luqman contributed on experimental setup, implementation on IoT-edge-cloud infrastructure, performance evaluation, data analysis, and preparation of figures and tables. M.U. Bokhari contributed in dataset preprocessing, literature review, result validation, proofreading, and refining the manuscript for final submission. B. Hanafi contributed significantly to the revision of the manuscript including analysis of reviewer feedback, enhancement of discussion in methodology section and overall improvement of writing quality. All authors have read and approved the final version of the manuscript.

Funding

No funding available.

Data availability

The data set generated during and/or analyzed during the current study is available in the GitHub repository, <https://github.com/DhilipSanjay/Detection-of-Sensitive-Data-Exposure-in-Images.git>.

Code availability

The code generated during and/or analyzed during the current study is available from the corresponding author upon reasonable requests.

Declarations

Ethics approval and consent to participate

This study utilized two types of data sources: (i) a publicly available open-source dataset obtained from a GitHub repository containing images with sensitive personal information (e.g., Aadhaar card images), and (ii) standard benchmark images commonly used in image processing research. The authors did not collect or generate any original personal data. All sensitive data were used strictly for academic research purposes without any attempt at individual identification. As all datasets used are publicly accessible and no direct human participation was involved, formal ethical approval was not required.

Consent for publication

Not applicable. No new personal or sensitive data were generated or disclosed by the authors.

Informed consent

Not applicable. The study did not involve direct interaction with human participants, and all data were obtained from publicly available sources.

Competing interests

The authors declare no competing interests.

Received: 13 August 2025 / Accepted: 13 January 2026

Published online: 01 February 2026

References

1. Chatterjee S, Kar AK, Gupta MP. Success of IoT in smart cities of india: an empirical analysis. *Gov Inf Q*. 2018;35(3):349–61. <https://doi.org/10.1016/j.giq.2018.05.002>.
2. Gubbi J, Buyya R, Marusic S, Palaniswami M, Internet of things (IoT): A vision, architectural elements, and future directions. *Futur Gener Comput Syst*. 2013.
3. De Mulder Y, Roelse P, Preneel B. Cryptanalysis of the Xiao - Lai white-box AES implementation. *Lect Notes Comput Sci (including Subser Lect Notes Artif Intell Lect Notes Bioinformatics)*. 2013;7707:34–49. https://doi.org/10.1007/978-3-642-35999-6_3. LNCS.
4. Hell M, Johansson T, Maximov A, Meier W. A stream cipher proposal: Grain-128. *IEEE Int Symp Inf Theory - Proc*. 2006;1614–8. <https://doi.org/10.1109/ISIT.2006.261549>.
5. Hell M, Johansson T, Meier W. Grain: A stream cipher for constrained environments. *Int J Wirel Mob Comput*. 2007;2(1):86–93. <https://doi.org/10.1504/IJWMC.2007.013798>.
6. Hamann M, Krause M, Meier W. LIZARD – A lightweight stream cipher for Power-constrained devices. *IACR Trans Symmetric Cryptol*. 2017;45–79. <https://doi.org/10.46586/tosc.v2017.i1.45-79>.
7. Boesgaard M, Vesterager M, Christensen T, Zenner E. The stream cipher rabbit. *Fast Softw Encryption*. 2003;10th Int Work FSE 2003(4986 LNCS):69–83.
8. De Cannière C. TRIVIUM: A stream cipher construction inspired by block cipher design principles. *Lect Notes Comput Sci (including Subser Lect Notes Artif Intell Lect Notes Bioinformatics)*. 2006;4176 LNCS:171–86. https://doi.org/10.1007/11836810_13.
9. Bernstein DJ. The salsa20 family of stream ciphers. *Lect Notes Comput Sci (including Subser Lect Notes Artif Intell Lect Notes Bioinformatics)*. 2008;4986 LNCS:84–97. https://doi.org/10.1007/978-3-540-68351-3_8.
10. Mishra Z, Acharya B. High throughput and low area architectures of secure IoT algorithm for medical image encryption. *J Inf Secur Appl*. 2020;53. <https://doi.org/10.1016/j.jisa.2020.102533>.
11. Gafsi M, Abbassi N, Hajjaji MA, Malek J, Mtibaa A. Improved chaos-based cryptosystem for medical image encryption and decryption. *Sci. Program*, vol. 2020, 2020. <https://doi.org/10.1155/2020/6612390>
12. Aljumah A. Assessment of machine learning techniques in IoT-based architecture for the monitoring and prediction of COVID-19. *Electron*. 2021;10(15). <https://doi.org/10.3390/electronics10151834>.
13. Mahmood MT, Ahmed SRA, Ahmed MRA. Using machine learning to secure IOT systems. *4th Int Symp Multidiscip Stud Innov Technol ISMSIT 2020 - Proc*. 2020. <https://doi.org/10.1109/ISMSIT50672.2020.9254304>.
14. Citić I, Peraković D, Periša M, Gupta B. Ensemble machine learning approach for classification of IoT devices in smart home. *Int J Mach Learn Cybern*. 2021;12(11):3179–202. <https://doi.org/10.1007/s13042-020-01241-0>.
15. Islam MR, Kabir MM, Mridha MF, Alfarhood S, Safran M, Che D. Deep Learning-Based IoT system for remote monitoring and early detection of health issues in Real-Time. *Sensors*. 2023;23(11). <https://doi.org/10.3390/s23115204>.
16. Shaikh R, Sasikumar M. Data classification for achieving security in cloud computing. *Procedia Comput Sci*. 2015;45:493–8. <https://doi.org/10.1016/j.procs.2015.03.087>.
17. Kethineni K, Gera P. IoT-Based Privacy-Preserving anomaly detection model for smart agriculture. *Systems*. 2023;11(6). <https://doi.org/10.3390/systems11060304>.
18. Moso JC, Cormier S, de Runz C, Fouchal H, Wandeto JM. Anomaly detection on data streams for smart agriculture. *Agric*. 2021;11(11). <https://doi.org/10.3390/agriculture11111083>.
19. Chaudhary N, Shahi TB, Neupane A. Secure image encryption using chaotic, hybrid chaotic and block cipher approach. *J Imaging*. 2022;8(6). <https://doi.org/10.3390/jimaging8060167>.
20. Kalso A, Ghebleh M. An efficient and robust image encryption scheme for medical applications. *Commun Nonlinear Sci Numer Simul*. 2015;24:1–3. <https://doi.org/10.1016/j.cnsns.2014.12.005>.
21. Kumar M, Iqbal A, Kumar P. A new RGB image encryption algorithm based on DNA encoding and elliptic curve Diffie-Hellman cryptography. *Sig Process*. 2016;125:187–202. <https://doi.org/10.1016/j.sigpro.2016.01.017>.
22. Hasan MK, et al. Lightweight encryption technique to enhance medical image security on internet of medical things applications. *IEEE Access*. 2021;9:47731–42. <https://doi.org/10.1109/ACCESS.2021.3061710>.
23. Wu Y, Zhang L, Berretti S, Wan S. Medical image encryption by Content-Aware DNA computing for secure healthcare. *IEEE Trans Ind Inf*. 2023;19(2):2089–98. <https://doi.org/10.1109/TII.2022.3194590>.
24. Liu J, Ma Y, Li S, Lian J, Zhang X. A new simple chaotic system and its application in medical image encryption. *Multimed Tools Appl*. 2018;77:22787–808. <https://doi.org/10.1007/s11042-017-5534-8>.
25. Gupta MR, Singh VP, Gupta KK, Shukla PK. An efficient image encryption technique based on two-level security for internet of things. *Multimed Tools Appl*. 2023;82(4):5091–111. <https://doi.org/10.1007/s11042-022-12169-8>.
26. Roy S, Rawat U, Sareen HA, Nayak SK. IECA: an efficient IoT friendly image encryption technique using programmable cellular automata. *J Ambient Intell Humaniz Comput*. 2020;11(11):5083–102. <https://doi.org/10.1007/s12652-020-01813-6>.
27. Dhopavkar TA, Nayak SK, Roy S. IETD: a novel image encryption technique using Tinkerbell map and duffing map for IoT applications. *Multimed Tools Appl*. 2022;81:43189–228. <https://doi.org/10.1007/s11042-022-13162-x>.
28. Liu W, Wang H, Chen Y, Sun K. A new image encryption scheme based on block compressive sensing and chaotic laser system for IoT. *Eur Phys J Plus*. 2024;139(6). <https://doi.org/10.1140/epjp/s13360-024-05221-z>.
29. Afify YM, Sharkawy NH, Gad W, Badr N. A new dynamic DNA-coding model for gray-scale image encryption. *Complex Intell Syst*. 2024;10(1):745–61. <https://doi.org/10.1007/s40747-023-01187-0>.
30. Alexan W, Shabasy NHE, Ehab N, Maher EA. A secure and efficient image encryption scheme based on chaotic systems and nonlinear transformations. *Sci Rep*. 2025;15(1). <https://doi.org/10.1038/s41598-025-15794-z>.

31. Lu W, Jin C, Wang J, Liu X, Liu J, Zhai Z. A novel image encryption scheme using 3D chaotic maps with Josephus permutation and dynamic diffusion. *J King Saud Univ - Comput Inf Sci*. 2025;37(8). <https://doi.org/10.1007/s44443-025-00284-z>.
32. Wen H, Lin Y, Kang S, Zhang X, Zou K. Secure image encryption algorithm using chaos-based block permutation and weighted bit planes chain diffusion. *iScience*. 2024;27(1). <https://doi.org/10.1016/j.isci.2023.108610>.
33. Wang M, Fu X, Yan X, Teng L. A new Chaos-Based image encryption algorithm based on discrete fourier transform and improved Joseph traversal. *Mathematics*. 2024;12(5). <https://doi.org/10.3390/math12050638>.
34. Asiri F, et al. Enhancing medical image privacy in IoT with bit-plane level encryption using chaotic map. *Front Comput Neurosci*. 2025;19. <https://doi.org/10.3389/fncom.2025.1591972>.
35. Bokhari MU, Shallal QM, Tamandani YK. Reducing the required time and power for data encryption and decryption using K-NN machine learning. *IETE J Res*. 2019;65(2):227–35. <https://doi.org/10.1080/03772063.2017.1419835>.
36. Wu J, Ping L, Ge X, Ya W, Fu J. Cloud storage as the infrastructure of cloud computing. *Proc - 2010 Int Conf Intell Comput Cogn Inf ICICCI 2010*. 2010;380–3. <https://doi.org/10.1109/ICICCI.2010.119>.
37. Zhang Q, Guo L, Wei X. Image encryption using DNA addition combining with chaotic maps. *Math Comput Model*. 2010;52:11–2. <https://doi.org/10.1016/j.mcm.2010.06.005>.
38. Alrubaie AH, Khodher MAA, Abdulameer AT. Image encryption based on 2DNA encoding and chaotic 2D logistic map. *J Eng Appl Sci*. 2023;70(1). <https://doi.org/10.1186/s44147-023-00228-2>.
39. Gabr M, et al. Application of DNA Coding, the Lorenz differential equations and a variation of the logistic map in a Multi-Stage cryptosystem. *Symmetry (Basel)*. 2022;14(12). <https://doi.org/10.3390/sym14122559>.
40. Midoun MA, Wang X, Talhaoui MZ. A sensitive dynamic mutual encryption system based on a new 1D chaotic map. *Opt Lasers Eng*. 2021;139. <https://doi.org/10.1016/j.optlaseng.2020.106485>.
41. Laiphrakpam DS, Khumanthem MS. Medical image encryption based on improved elgamal encryption technique. *Optik (Stuttg)*. 2017;147:88–102. <https://doi.org/10.1016/j.ijleo.2017.08.028>.
42. Ding Y, et al. DeepEDN: A Deep-Learning-Based image encryption and decryption network for internet of medical things. *IEEE Internet Things J*. 2021;8(3):1504–18. <https://doi.org/10.1109/JIOT.2020.3012452>.
43. Alvarez G, Li S. Some basic cryptographic requirements for chaos-based cryptosystems. *Int J Bifurc Chaos*. 2006;16:2129–51. <https://doi.org/10.1142/S0218127406015970>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.